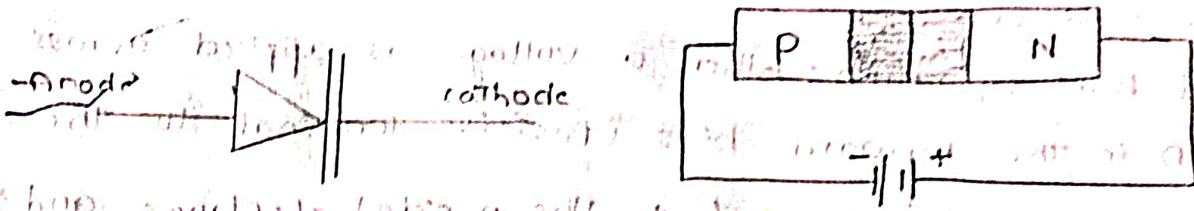


UNIT-II

①

1. Explain the operation of varactor diode with neat diagram.

Ans:



To know the varactor diode working principle, we must know the function of capacitor and capacitance. Let us know

Now, consider the capacitor that comprises of two plates separated by an insulator. Varactor diode is a special type of diode which uses transition capacitance property i.e. voltage variable capacitance. These are also called as varicap, VVC (Voltage Variable capacitance) or tuning diodes.

When a reverse voltage is applied to a PN Junction, the holes in the p-region are attracted to the anode terminal and electrons in the n-region are attracted to the cathode terminal creating a region where there is little current.

The depletion region increases as reverse voltage across it increases; and since capacitance varies inversely as dielectric thickness, the junction capacitance will decrease as the voltage across the PN Junction increases. So, by varying the reverse voltage across a PN Junction the junction capacitance can be varied.

2. Explain the operation of LED with neat diagram.

Ans: Operation of LED:

Forward Bias Operation: When a voltage is applied across the LED in the forward bias (positive terminal to the P-side and negative terminal to the n-side), electrons and holes are injected into the junction.

⇒ Electrons move from the n-side to the p-side and holes move from the p-side to n-side.

Recombination: At the junction, electrons recombine with holes. The wave length of the emitted light depends on the energy gap of the semiconductor material.



Fig a: LED

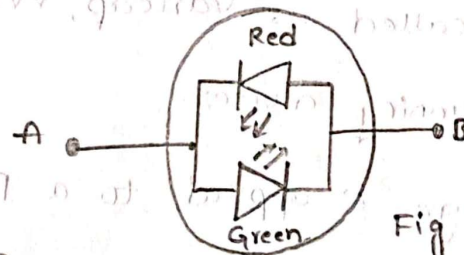


Fig b: The light Emitting Diode.

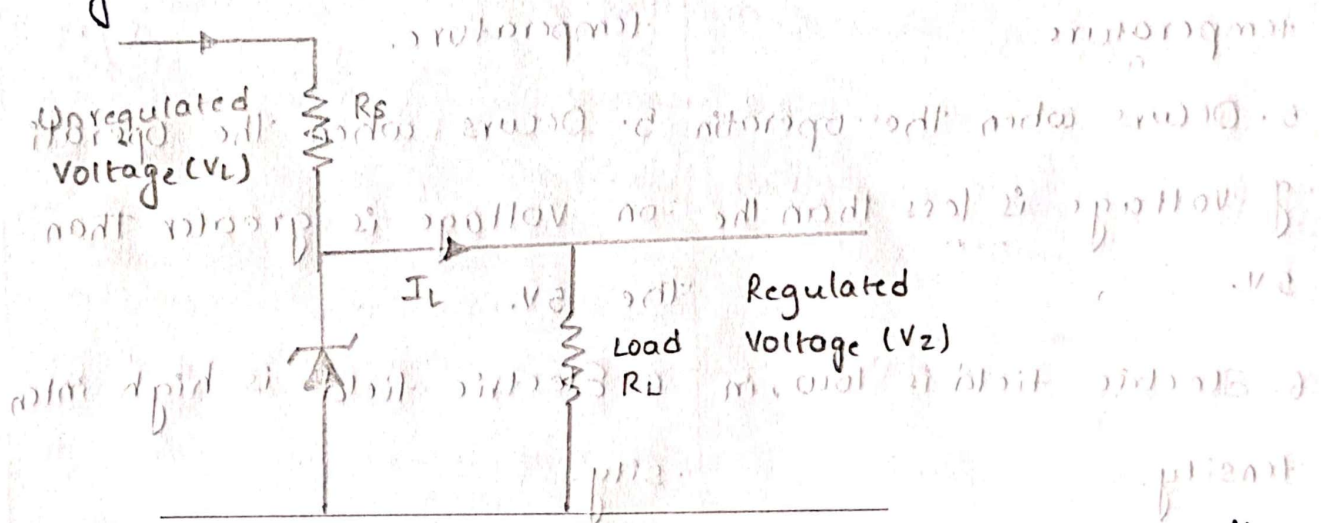
Efficiency: LED's are highly Efficient because most of the energy is converted into light, with minimal heat generation compared to incandescent bulbs.

Polarization: LED's are polarized devices, meaning they only emit light when the current flows in the correct direction (F.B). In Reverse Bias, the LED does not conduct and does not emit light.

3. Explain how the Zener diode is used for regulation purpose.

Ans: Zener Diode as a voltage Regulator:

There is a series resistor connected to the circuit in order to limit the current into the diode. It is connected to limit the current connected to the positive terminal of the d.c. It works in such a way the reverse-biased can also work in breakdown conditions. We do not use ordinary Junction diode because the low power rating diode can get damaged when we apply reverse bias above its breakdown voltage. When the minimum input voltage and the maximum load current is applied, the Zener diode current should always be minimum.



Since the input voltage and the required output voltage is known, it is easier to choose a Zener diode with a voltage approximately equal to the load voltage i.e. $V_Z = V_L$. The Zener diode is used as a shunt voltage regulator for regulating voltage across small loads.

4. Distinguish between -Avalanche and Zener mechanism of a diode

Ans.

Avalanche Diode	Zener Breakdown
1. It occurs at heavily doped PN Junctions	1. It occurs at lightly doped PN Junction.
2. The width of depletion region is wide	2. The depletion layers are narrow.
3. The junction is doped lightly	3. The junction is doped heavily.
4. Breakdown voltage is directly proportional to temperature	4. Breakdown voltage is inversely proportional to the temperature
5. Occurs when the operating voltage is less than the 5V.	5. Occurs when the operating voltage is greater than the 5V.
6. Electric field is low, in intensity.	6. Electric field is high intensity.
7. The voltage varies after the Breakdown.	7. There is no Effect of breakdown on voltage
8. The temperature coefficient is positive	8. The temperature coefficient is Negative

5. Explain the working of Tunnel diode and Zener diode and its $V-I$ characteristics and what is the sufficient condition for tunnelling.

Ans.

Working of Tunnel Diode:

Tunneling effect: When a semiconductor is heavily doped with impurities, electrons can tunnel directly from the n-side to the p-type. This is because electrons have wavelike properties in the quantum mechanical world.

Negative resistance region: As voltage increases tunnel through the P-N junction barrier and fill electron states in the conduction band on the N-side.

Transition point: When voltage increases past a fixed transition point, the diode operates as a normal diode, with electrons travelling across the P-N junction by conduction.

Zener Diode Working

High-level impurities are added to a Zener diode to make it more conductive and thus the Zener diodes can easily conduct electricity compared to other P-n junction diodes.

In no biasing condition of the Zener diode, all the electrons accumulate in the valence band of the P-type semiconductor material and thus no current flow occurs through the diode.

In reverse bias conditions, if the zener voltage is equal to the supplied voltage, the diode conducts electricity in the direction of reverse bias.

V-I characteristics of Tunnel diode:

Some of the characteristics are Negative resistance, low inductance and capacitance, ultra high-speed switching device.

V-I characteristics of Zener diode:

The V-I characteristics of a zener diode are described by many conditions. They are Forward bias, Reverse bias,

Breakdown Voltage.

What is the sufficient condition for tunnelling?

Tunneling in a Tunnel diode occurs when charge carriers pass through the junction without kinetic energy.

This happens when the voltage across the diode increases, causing electrons to overcome a potential barrier and tunnel through it.

6. With the circuit and necessary wave forms Explain the operation of. 1. half wave rectifier 2. full wave rectifier 3. Bridge rectifier.

Ans: Half wave rectifier: A half wave rectifier is defined as a type of rectifier that only allows one half-cycle of an AC voltage waveform to pass, blocking the other half-cycle. Half-wave rectifiers are used to convert AC voltage to DC voltage, and only require a single diode to construct.

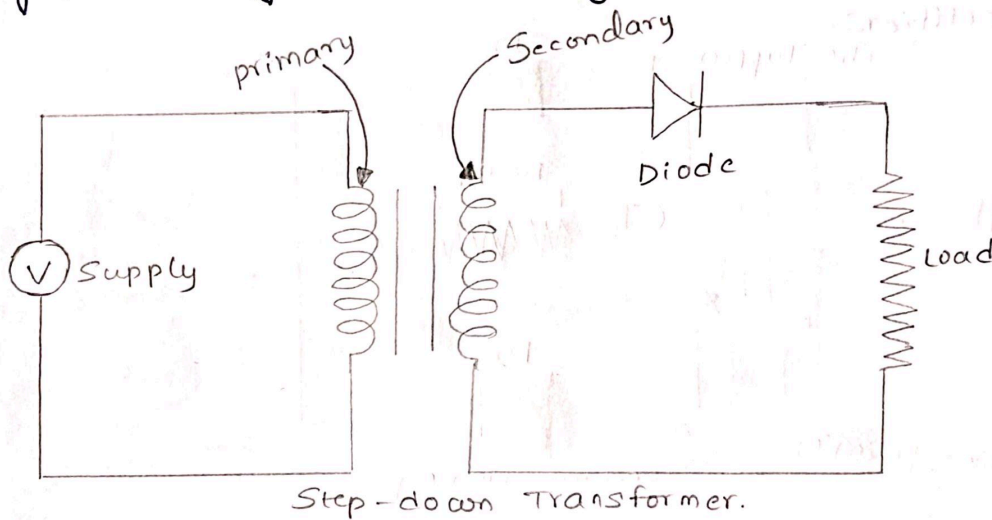
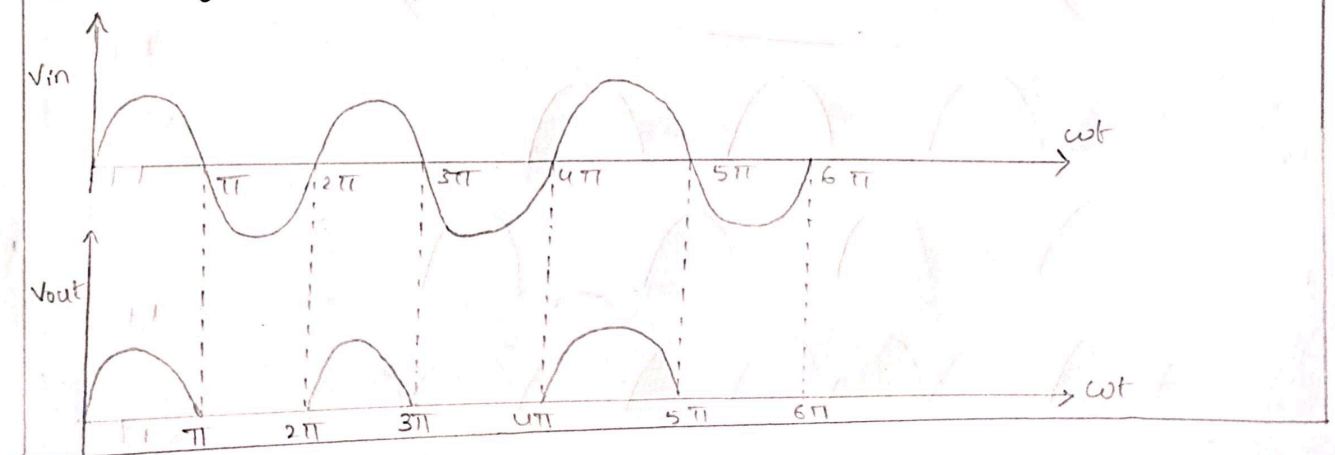
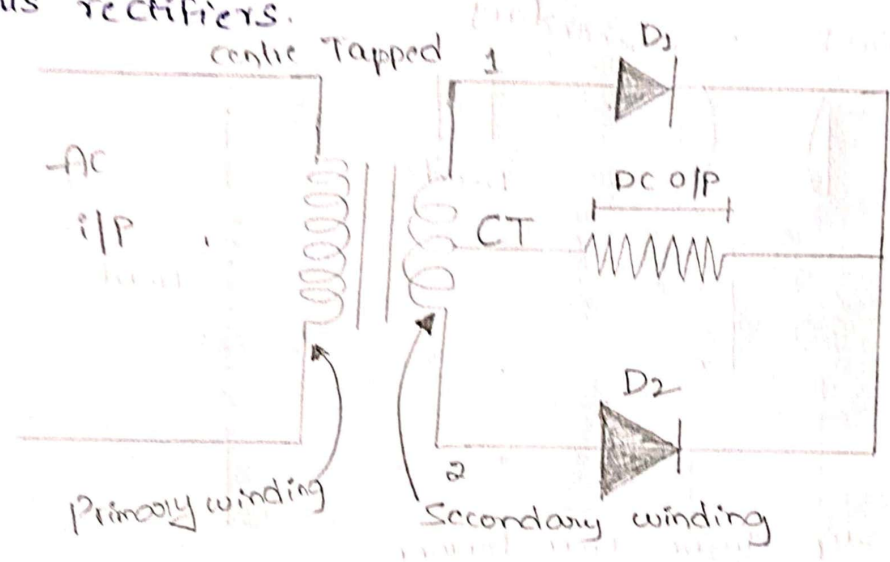


Fig a: half-wave rectifier

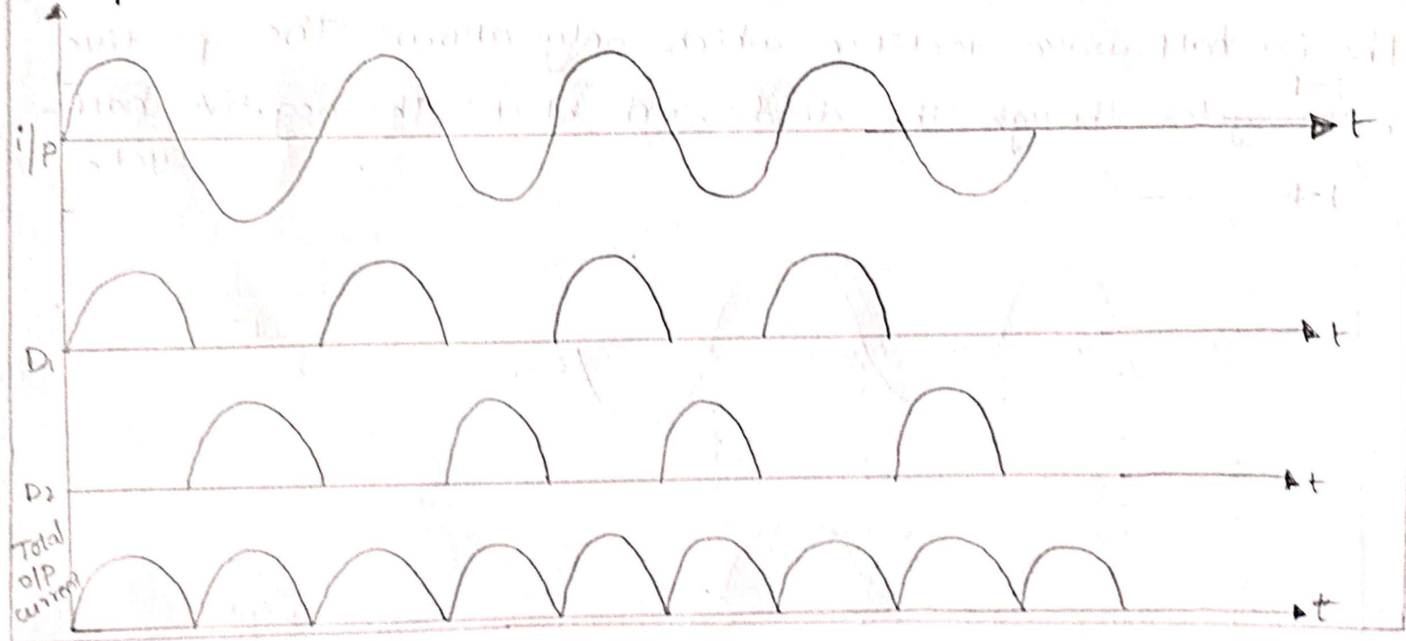
This is a half-wave rectifier which only allows the positive half-cycles through the diode, and blocks the negative half-cycle.



Full wave rectifier: -A full wave rectifier is defined as a type of rectifier that converts both halves of each cycle of an alternating wave (AC signal) into a pulsating DC signal. Full wave rectifiers are used to convert AC voltage to DC voltage requiring multiple diodes to construct. Full wave rectification is the process of AC to DC signal. A diode permits current in one direction only and blocks the current in the other direction. We use this principle to construct various rectifiers.



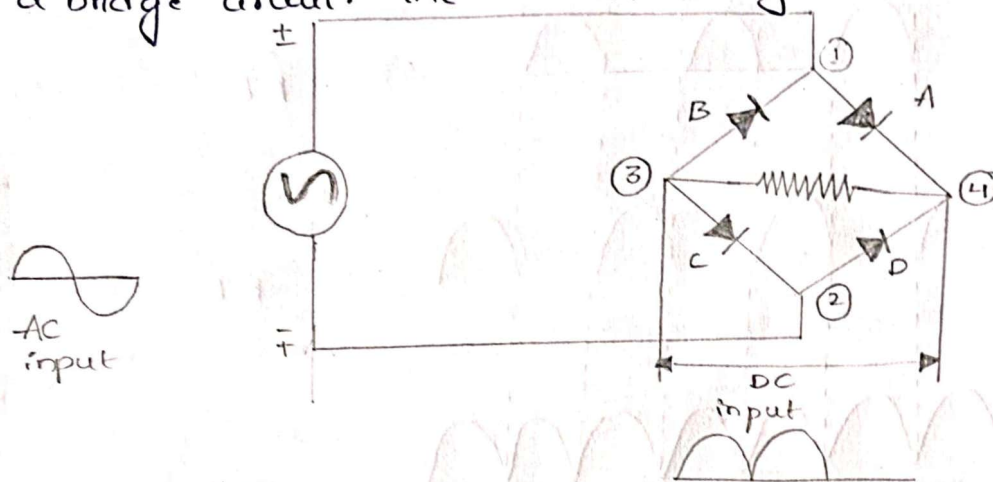
Output waveforms:



(9)

Full-wave Bridge Rectifier: A full wave bridge rectifier is a rectifier that will use four diodes or more than that in a bridge formation. A full wave bridge rectifier system consists of 1. Four diodes 2. Resistive Load.

We use the diodes, namely A, B, C and D, which form a bridge circuit. The circuit diagram is as follows.

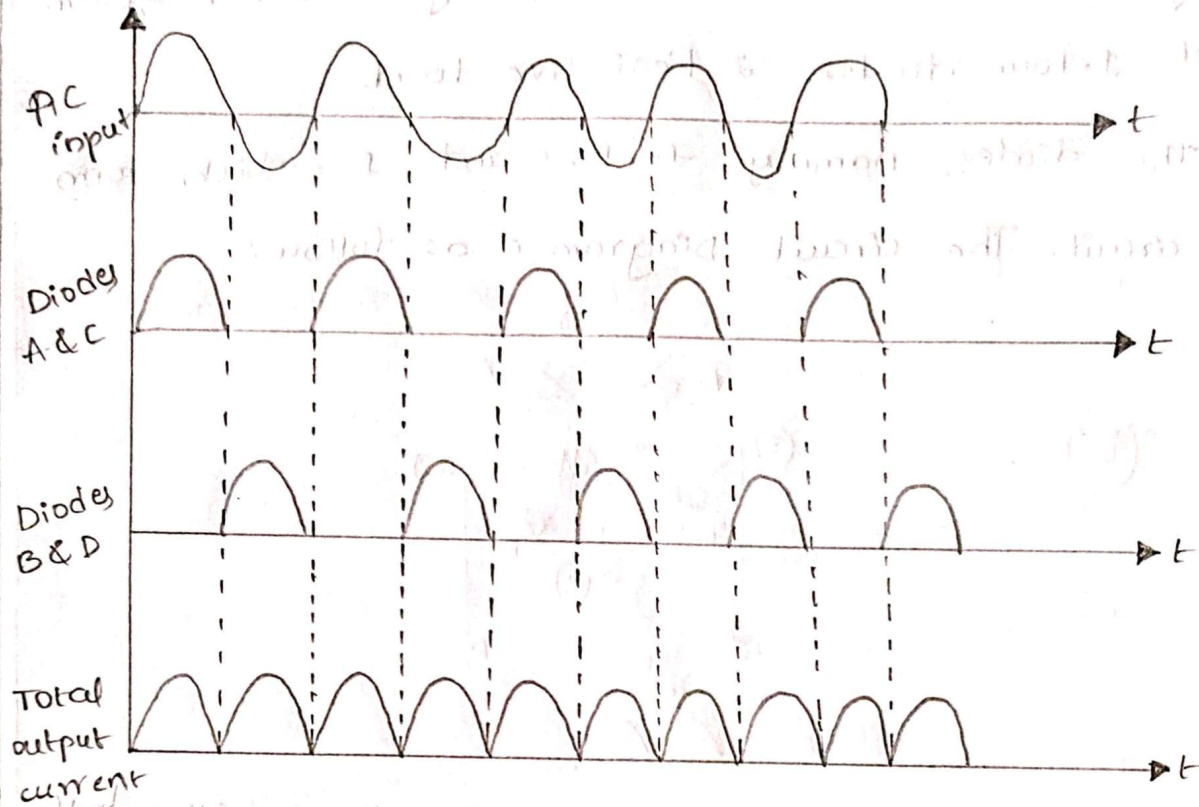


We apply an AC across the bridge. During the positive half cycle, terminal 1 becomes positive, and terminal 2 becomes negative. This will cause the diodes A and B to become forward-biased, and the current will flow through it. Meanwhile, diodes C and D will become reverse-biased and block current through them. The current will flow from 1 to 4 to 3 to 2.

During the Negative half cycle, terminal 1 will become negative, and terminal 2 will become positive.

This will cause the diodes B and D to become forward-biased and will allow current through them.

At the same time, diodes A and C will reverse-biased and will block the current through them. The current will flow from 2 to 4 to 3 to 1.



7. Explain the construction of photo Diode and working also.

Ans. The photo diode is a semiconductor p-n junction device whose region of operation is limited to the reverse biased region. The figure below shows the symbol of photodiode.

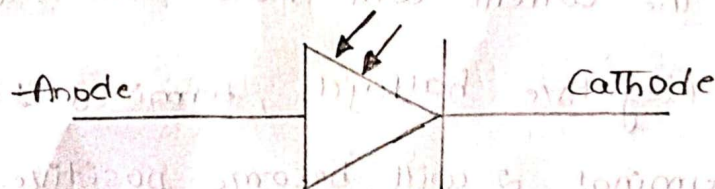
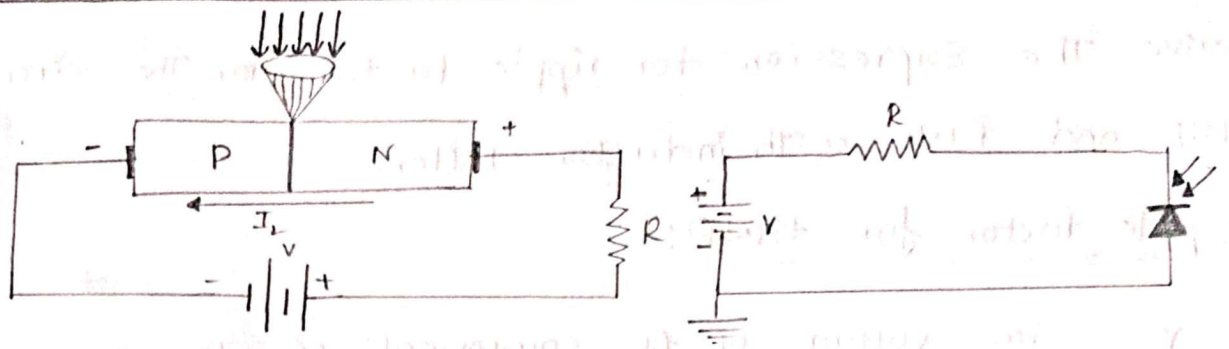


Fig: Symbol for photodiode



Basic Biasing Arrangement and construction photodiode

Working of Photodiode:

The working principle of a photodiode is, when a photon of of ample energy strikes the diodes, it makes a couple of an electron-hole. This mechanism is also called The inner photoelectric effect. If the absorption arises in the depletion region junction, then the carriers are removed from the junction by the inbuilt electric field of the depletion region. Therefore, holes in the region move toward the anode, and electrons move towards the cathode, and a photocurrent will be generated. The entire current through the diode is the sum of the absense of light and the photocurrent. So the absent current must be reduced to maximize the sensitivity of the device.

8. Derive The Expression for ripple factor for the circuit HWR and FWR with inductor filter.

Ans: Ripple factor for HWR:

$$r = \frac{\text{rms voltage of AC components of o/p}}{\text{Avg voltage of o/p}}$$

Total current I is

$$I = I_{ac} + I_{dc}$$

$$I_{ac} = I - I_{dc}$$

$$I_{ac} = I_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} I_{ac}^2 d(\omega t)}$$

$$= \left[\frac{1}{2\pi} \int_0^{2\pi} (I - I_{dc})^2 d(\omega t) \right]^{1/2}$$

$$= \left[\frac{1}{2\pi} \int_0^{2\pi} I^2 d(\omega t) + \int_0^{2\pi} I^2 dc \cdot d(\omega t) - \int_0^{2\pi} 2I \cdot I_{dc} d(\omega t) \right]^{1/2}$$

where

$$(I_{rms})^{1/2} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} I^2 d(\omega t)}$$

$$I_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} I^2 d(\omega t)}$$

$$= \left[I_{rms}^2 + I_{dc}^2 - 2I_{dc} \right]^{1/2}$$

$$I_{ac} = \left[I_{rms}^2 - I_{dc}^2 \right]^{1/2}$$

$$r = \frac{I_{ac}}{I_{dc}} = \frac{\left[I_{rms}^2 - I_{dc}^2 \right]^{1/2}}{I_{dc}}$$

$$r = \left[\frac{I_{rms}^2 - I_{dc}^2}{I_{dc}^2} \right]^{1/2} \Rightarrow \left[\frac{I_{rms}^2}{I_{dc}^2} - 1 \right]^{1/2}$$

$$\Rightarrow r = \sqrt{(F.F)^2 - 1}$$

$$\boxed{r = 1.21}$$

Ripple factor for FWR:

The o/p of fullwave rectifier contains both AC & DC components, the ripple factor measures the percentage of AC components in the rectified o/p.

$$r = \frac{\text{Rms value of AC components of o/p}}{\text{Avg. value of o/p.}}$$

Total current I is,

$$I = I_{ac} + I_{dc}$$

$$I = I - I_{dc}$$

$$r = \sqrt{(F.F)^2 - 1}$$

$$r = \sqrt{(1.11)^2 - 1}$$

$$\boxed{r = 0.48}$$

9. Explain about different types of clippers and clampers with neat circuit diagram.

Ans: There are 5 type of clippers. They are

1. Shunt diode positive clipper

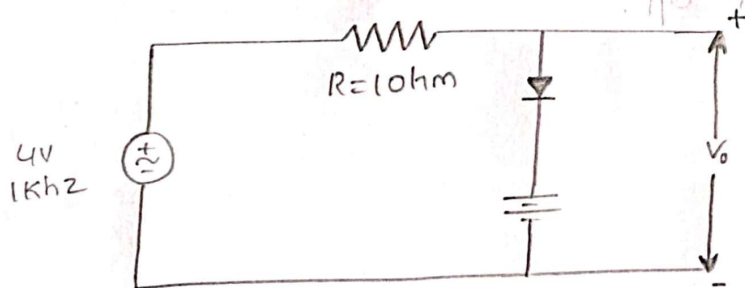
2. Shunt diode negative clipper

3. Series diode positive clipper

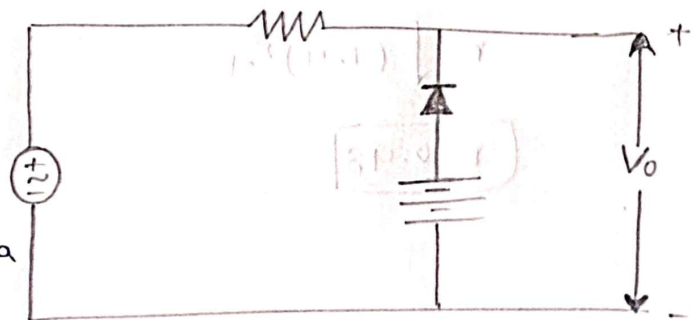
4. Series diode negative clipper

5. Two level clipper

1. Shunt diode positive clipper: A clipper circuit in which the diode is connected in shunt to the input signal and that attenuates the positive portions of the wave form.

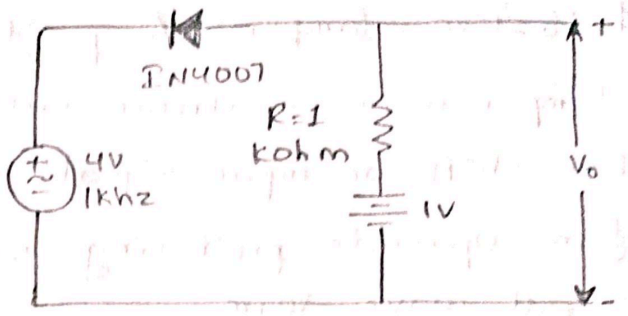


2. Shunt diode negative clipper: A clipper circuit in which the diode is connected in shunt to the input signal and that attenuates the negative portions of the wave forms.



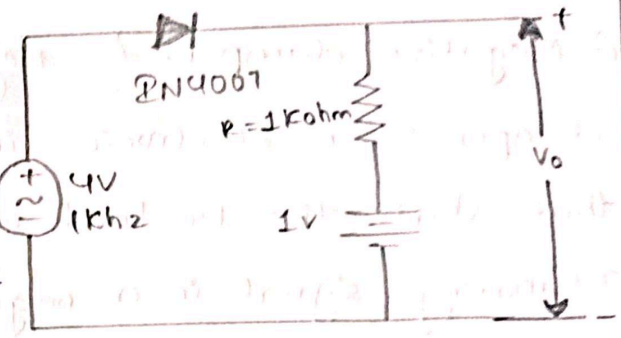
3. Series diode positive clipper:

This type of clipper has a diode connected in series with the input voltage signal to attenuate all the positive waveform portions.



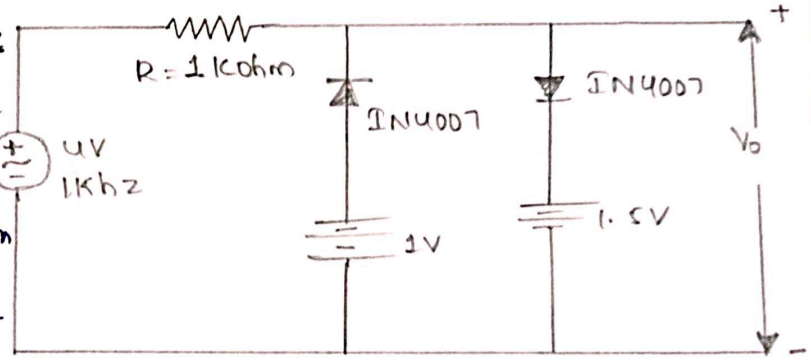
4. Series diode Negative clipper:

This type of clipper has a diode connected in series with the input voltage signal to attenuate all the Negative waveform portions.



5. Two level clipper:

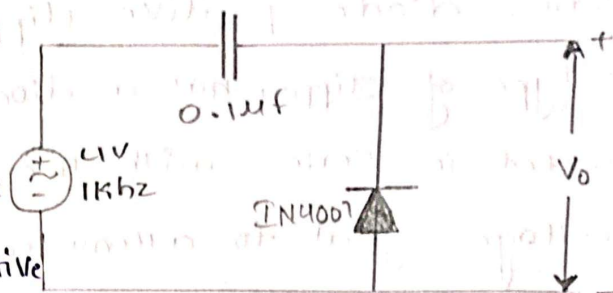
A two level clipper is an Electronic circuit that uses diodes and resistors to clip an input sine wave to two specific voltage levels.



There are two types of clammers: They are

1. Positive clamper
2. Negative clamper

1. Positive clamper: A positive clamper is an Electronic circuit that shifts an input signal's wave form upwards, producing a positive output wave form.



2. Negative clamper: A Negative clamper is an Electronic circuit that shifts the DC level of an Alternating signal in a negative direction, or below zero.

